

Supplementary material: Two Raman phonons measure the fixed edge charge that limits width scaling in monolayer semiconductor nanoribbon transistors

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(Dated: 29 July 2026)

S1. FIRST-PRINCIPLES CALCULATIONS

Setup. All electronic-structure calculations use the periodic module of PySCF¹ with structures built through the Atomic Simulation Environment.² We use the PBE functional,³ separable dual-space Gaussian pseudopotentials of the Goedecker-Teter-Hutter family^{4,5} in their PBE parameterisation (Mo and W with 14 valence electrons, S and Se with 6), and the DZVP-MOLOPT-SR Gaussian basis.⁶ The monolayer sits in a hexagonal cell whose c axis is 15 Å long, leaving about 12 Å of vacuum between periodic images, the density grid comes from a 60 Ha kinetic-energy cutoff, and the Brillouin zone is sampled on a $4 \times 4 \times 1$ mesh with a self-consistency threshold of 10^{-7} Ha. Band energies at high-symmetry points are obtained non-self-consistently from the converged density. Figure S1 shows the convergence behaviour. The total energy per atom is converged to better than 1 meV with respect to both the density cutoff and the vacuum thickness. With respect to the k -mesh it is not: going from $4 \times 4 \times 1$ to $6 \times 6 \times 1$ still moves the total energy by about 19 meV per atom. That absolute offset is largely common to every structure in a sweep and so largely cancels in the quantities we use, all of which are differences taken at fixed sampling: the elastic constant, the frozen-phonon force constants and their strain derivatives. We verified this directly by repeating the whole MoS₂ strain sweep and the A'_1 frozen-phonon calculation on a $6 \times 6 \times 1$ mesh, which gives $\gamma_{\text{LA}} = 2.24$ against 2.30, and, applying the same symmetric-displacement reduction at both meshes, $\gamma_{A'_1} = 0.62$ against 0.63. The residual few-per-cent sensitivity carries through to the interior strain of Section S2 at the same level, and not at all to the edge charge, which that section shows is independent of γ_{LA} . The sampling mesh does not contain the K point, which is why band energies are taken non-self-consistently at the high-symmetry points from the converged density. The absolute gap is a PBE gap and underestimates the quasiparticle gap; this does not affect any result reported, because only derivatives with respect to strain and differences within one calculation are used.

Strain sweep and elastic constants. The in-plane lattice constant is scaled by $1 + \varepsilon$ with $\varepsilon \in \{-2, -1, 0, +1, +2\}\%$, and the chalcogen height by $1 - \nu_{\perp}\varepsilon$ with $\nu_{\perp} = 0.25$; the

chalcogen height at zero strain is first relaxed by a three-point parabolic scan. For an isotropic two-dimensional sheet under equal biaxial strain the elastic energy density is $u = (C_{11} + C_{12})\varepsilon^2$, so

$$C_{11} + C_{12} = \frac{1}{2A_0} \frac{d^2 E}{d\varepsilon^2}, \quad (\text{S1})$$

with A_0 the cell area. We take $C_{12} \simeq 0.25 C_{11}$ to extract C_{11} ; this enters only the zone-edge acoustic frequency estimate.

Because the zone-edge longitudinal acoustic frequency obeys $\omega_{\text{LA}}(M) \propto \sqrt{C_{11}}$ once the area scaling of the Brillouin zone is included, its Grüneisen parameter is

$$\gamma_{\text{LA}} = -\frac{1}{2} \frac{d \ln C_{11}}{d \ln A}. \quad (\text{S2})$$

We evaluate this without ambiguity by fitting a local three-point quadratic to $E(\varepsilon)$ centred at $\varepsilon = \pm 1\%$, dividing by the strained cell area at that point, and differencing the logarithms. For MoS₂ this gives $\gamma_{\text{LA}} = 2.30$. Section S2 shows that the edge assignment is independent of this value.

Frozen phonons. The two Raman-active zone-centre modes are obtained by symmetry-adapted displacements. For the in-plane E' mode the metal and the chalcogen pair move oppositely with zero net momentum, $m_M u_M = -2m_X u_X$, so the mass-weighted normal coordinate satisfies $Q^2 = 2m_X u_X^2 (1 + 2m_X/m_M)$, and the force constant is taken from the symmetric second difference

$$\omega^2 = \frac{E(+u) + E(-u) - 2E(0)}{Q^2}, \quad (\text{S3})$$

with $u = 0.05$ Å. Using both signs matters because the chalcogen height is relaxed only at zero strain, so at finite strain the structure carries a small residual force along the mode coordinate, and a one-sided difference would alias that force into the extracted Grüneisen parameter.

The A'_1 mode is handled differently, and more directly. Its normal coordinate is the chalcogen half-thickness itself, with $Q^2 = 2m_X z^2$, so a three-point parabolic scan of that single coordinate returns both the relaxed geometry, from the position of the minimum, and the force constant, from the curvature there. Evaluating the curvature at the relaxed point removes the residual-force problem altogether, so this is the route used for every $\omega_{A'_1}$ and $\gamma_{A'_1}$ reported here, with a scan half-width of 0.04 Å. The harmonic assumption is checked by repeating the MoS₂

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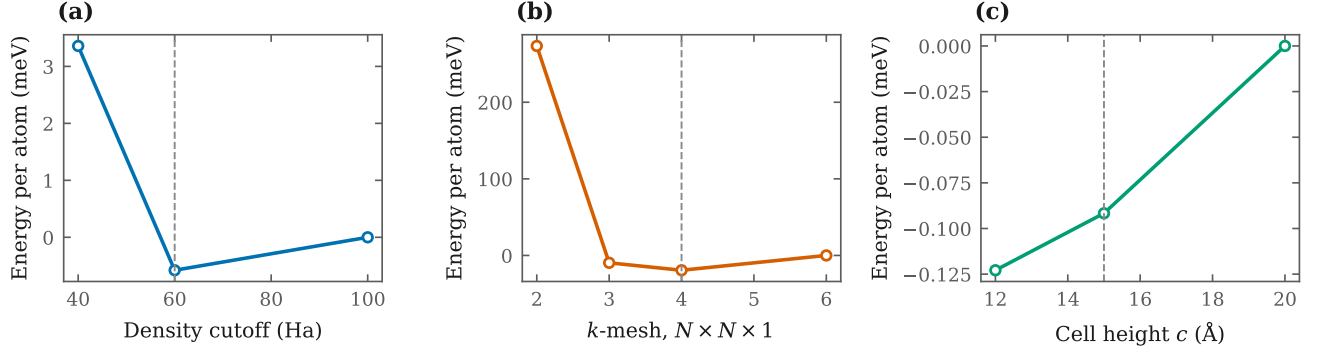


FIG. S1. Convergence for monolayer MoS₂. (a) Energy per atom against the density-grid cutoff, referenced to 100 Ha. (b) Energy per atom against k -point sampling, referenced to a $6 \times 6 \times 1$ mesh. (c) Energy per atom against the cell height c , that is the repeat length along the axis normal to the layer; the vacuum gap is c less the layer thickness of about 3.1 Å. Dashed lines mark the production settings.

symmetric-displacement calculation at $u = 0.025, 0.05$ and 0.075 Å, over which $\omega_{A'_1}$ moves by 0.5 cm^{-1} .

Repeating at $\varepsilon = \pm 1\%$ gives the mode Grüneisen parameters, $\gamma_i = -(1/2\omega_i) d\omega_i/d\varepsilon$, where the factor of two converts linear biaxial strain to areal strain. Figure S2 shows the underlying energy curves and compares the computed zone-centre frequencies with experiment; Table S1 collects the constants. Measured mode Grüneisen parameters across the full resonant Raman band set of MoS₂ are available for comparison.⁷ Absolute frequencies carry the usual few-percent PBE error, so the forward model uses measured zone-centre frequencies as the anchor^{8–10} and the computed Grüneisen parameters as the strain response.

S2. SEPARATING CHARGE FROM STRAIN

The two observables are the shifts of the first-order A'_1 mode and of the disorder-activated 2LA(M) overtone relative to an unperturbed reference,

$$\begin{pmatrix} \Delta\omega_{A'_1} \\ \Delta\omega_{2LA} \end{pmatrix} = \begin{pmatrix} -2\gamma_{A'_1}\omega_{A'_1}^0/100 & \lambda_{A'_1} \\ -2\gamma_{2LA}\omega_{2LA}^0/100 & 0 \end{pmatrix} \begin{pmatrix} \varepsilon \\ n \end{pmatrix}, \quad (\text{S4})$$

with ε in percent and n in units of 10^{13} cm^{-2} . The zero in the lower right is the physical content of the method. The renormalisation of the out-of-plane optical mode is a measured, symmetry selective coupling to the K -valley electron density;¹¹ the corresponding coefficient for the zone-edge acoustic overtone has not been measured, and setting it to zero is an assumption, not a citation. It is bounded below rather than asserted. The system is solved as a regularised least squares problem with measurement standard deviations of 0.15 and 0.30 cm^{-1} on the two channels and a broad prior, and the reported error bars are the square roots of the diagonal of the posterior covariance.

Because $\Delta\omega_{2LA} = 0$ at the ribbon edge within the measurement uncertainty, the second row of Eq. (S4) pins ε near zero for *any* non-zero value of γ_{2LA} , and the entire A'_1 shift is carried by the second column. This is why the recovered edge charge in Fig. 3(a) of the main text is flat across the whole scanned range of γ_{2LA} : the spread is 1% over $0.4 \leq \gamma_{2LA} \leq 2.6$, a range that brackets the plausible values for MoS₂ shaded in that panel. Only the interior inhomogeneity, where the overtone does shift, depends on γ_{2LA} . For the same reason the recovered edge charge is independent of $\gamma_{A'_1}$: with ε pinned at zero the first row of Eq. (S4) reduces to $n = \Delta\omega_{A'_1}/\lambda_{A'_1}$. Adopting the measured optical Grüneisen parameter of MoS₂, $\gamma_{A'_1} = 0.31$, in place of the calculated 0.63 therefore leaves the edge density unchanged to four significant figures, and leaves the interior strain unchanged as well, because that is pinned by the 2LA(M) row, which contains no $\gamma_{A'_1}$. What it does change is the small residual charge assigned to the interior spot, which is consistent with zero either way.

A second assumption is that the overtone does not renormalise with carrier density. No gated measurement of that coefficient exists, so we bound it rather than assume it: setting λ_{2LA} to a fraction of the A'_1 coefficient scaled by the frequency ratio and repeating the solve, a fraction of one half changes the recovered edge density by 16%.

The dominant systematic uncertainty is therefore neither γ_{2LA} nor λ_{2LA} but the A'_1 doping coefficient $\lambda_{A'_1} = -2.2 \text{ cm}^{-1}$ per 10^{13} cm^{-2} , a measured quantity with roughly 15% uncertainty, to which the recovered density is inversely proportional. The second is the edge width w_e used to convert an areal to a line density. Tip-enhanced Raman resolves about 10 nm and nano-optical studies place the electronic edge region of MoS₂ between about 5 and 20 nm,^{12,13} so we take $w_e = 10 \text{ nm}$. The excess areal density n becomes a line density $\sigma_e = n w_e$ along each edge, which is the quantity that enters Eq. (1) of the main text; varying w_e over the range above moves

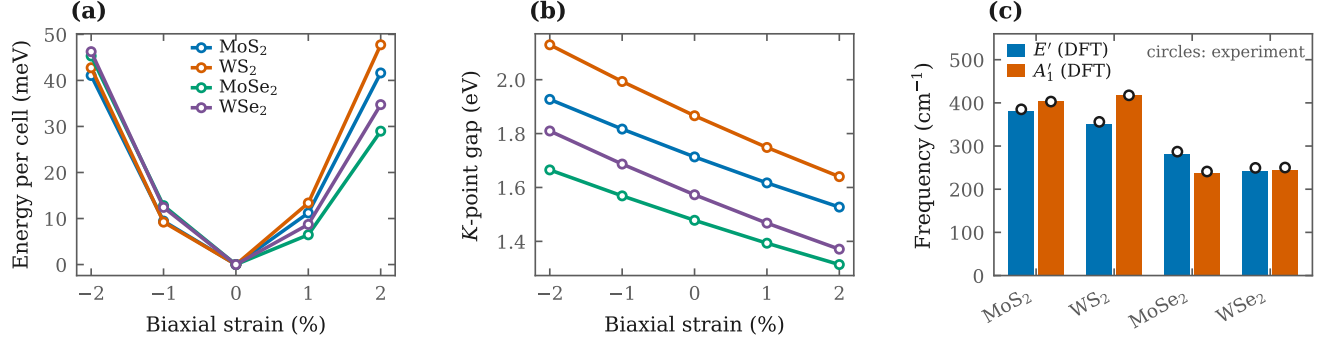


FIG. S2. (a) Total energy against biaxial strain, referenced to zero strain. (b) K -point gap against biaxial strain. (c) Frozen-phonon zone-centre frequencies compared with measured values.

TABLE S1. First-principles constants computed in this work. a_0 is the in-plane lattice constant used ($\text{\AA}$), $\omega_{E'}$ and $\omega_{A_1'}$ are the frozen-phonon zone-centre frequencies (cm^{-1}), $\gamma_{E'}$ and $\gamma_{A_1'}$ the corresponding mode Grüneisen parameters, γ_{LA} the acoustic Grüneisen parameter obtained from the strain dependence of the elastic constant, and C_{2D} the in-plane elastic constant (N/m). The lattice constant is an input; every other entry is computed in this work and none is fitted.

Material	a_0	$\omega_{E'}$	$\omega_{A_1'}$	$\gamma_{E'}$	$\gamma_{A_1'}$	γ_{LA}	C_{2D}
MoS ₂	3.184	380	405	1.01	0.63	2.30	151
WS ₂	3.183	350	418	1.05	0.65	2.34	165
MoSe ₂	3.319	281	237	1.00	0.66	3.07	124
WSe ₂	3.322	243	245	1.07	0.68	3.15	135

the critical width on the high- κ stack from 15 to 21 nm and leaves the halo-limited branch unchanged.

Finally, the sign of the edge charge is environment dependent. The ribbons measured by tip-enhanced Raman were supported on gold, which donates electrons; nano-optical measurements of MoS₂ edges on oxide instead report local depletion.^{12,14} Only the magnitude is transferred to the electrostatic analysis.

S3. TRANSPORT AND NANORIBBON MODEL

Scattering channels. The wide-channel mobility combines three channels by Matthiessen's rule. The phonon-limited value is the published first-principles K -valley-dominated mobility.¹⁵ Neutral point defects are treated as short-range scatterers, for which the two-dimensional Fermi golden rule gives an energy-independent rate,

$$\frac{1}{\tau_{\text{pd}}} = \frac{m^* n_d U^2}{\hbar^3}, \quad \mu_{\text{pd}} = \frac{e \hbar^3}{m^{*2} n_d U^2}, \quad (\text{S5})$$

with U the areal integral of the defect potential. Charge at the dielectric interface is treated with a screened two-dimensional Coulomb potential, $V(q) = e^2 \exp(-2qd)/[2\epsilon_0 \epsilon_{\text{env}}(q + q_s)]$ with the Thomas-Fermi wavevector $q_s = m^* e^2 / (2\pi \epsilon_0 \epsilon_{\text{env}} \hbar^2)$, a setback $d =$

0.3 nm representing the van der Waals gap, and

$$\frac{1}{\tau_{\text{ci}}} = \frac{m^* n_{\text{it}}}{2\pi \hbar^3} \int_0^{2\pi} |V(q)|^2 (1 - \cos \theta) d\theta, \quad (\text{S6})$$

with $q = 2k_F \sin(\theta/2)$. Figure S3(a) shows the decomposition, and Fig. S3(b) the width dependence of the ribbon mobility for four halo widths. Two constants are calibrated once and then held for every result: $U = 0.285 \text{ eV nm}^2$, fixed against the first-principles quantum transport result for disordered WS₂,¹⁶ and the interface charge density n_{it} of each gate stack, fixed against the measured field-effect mobility of a clean monolayer on SiO₂ and the measured wide-channel on-current on the high- κ stack. Everything else is computed or taken from a published measurement, as listed in Table S3.

Nanoribbon layer. Three effects are added. Diffuse edge scattering in the two-dimensional Fuchs-Sondheimer limit gives $\mu_{\text{edge}} = eW/(m^* \bar{v})$ with $\bar{v} = \sqrt{\pi k_B T / 2m^*}$; at the widths of interest this channel is weak, consistent with the measured absence of mobility degradation from 850 down to 75 nm.¹⁷ The patterning damage halo is treated as two parallel conductors,

$$\mu(W) = \frac{(W - 2w_h)\mu_{\text{int}} + 2w_h\mu_{\text{dam}}}{W}, \quad (\text{S7})$$

with μ_{int} and μ_{dam} evaluated at the bulk and elevated defect densities. Fixed edge charge shifts the threshold by Eq. (1) of the main text, reducing the gate overdrive,

TABLE S2. Inversion of the published atomic-layer-deposited and sputtered MoS₂ films. R_{sh} and R_{LA} are the two disorder-activated Raman ratios, n_{d} (cm⁻²) and L_{D} (nm) the inverted defect density and mean inter-defect distance, and the last two columns the measured and point-defect-limited mobility (cm²V⁻¹s⁻¹).

Film	R_{sh}	R_{LA}	n_{d}	L_{D}	μ_{meas}	μ_{ceiling}
ALD as-grown	0.325	0.043	7.3×10^{12}	3.7	0.16	29
ALD annealed	0.230	0.032	5.4×10^{12}	4.3	0.44	33
sputtered as-grown	0.687	0.128	2.2×10^{13}	2.1	0.03	14
sputtered annealed	0.507	0.087	1.5×10^{13}	2.6	—	19

TABLE S3. Provenance of every group of constants used in this work. Entries marked “this work” are computed here from first principles; “calibrated once” means the value was fixed a single time against the cited published result and then held for every number reported; “geometry” means a nominal device dimension; “assumed” marks a modelling choice stated in the text; and “not used” marks a constant that is present in the code but multiplies zero in every result reported here.

Constants	Origin	Source
BANDS	literature	Jin et al., Phys. Rev. B 90, 045422 (2014)
MU_PHONON	literature	Jin et al., Phys. Rev. B 90, 045422 (2014)
PHONON	literature	Mignuzzi 2015, Berkdemir 2013, Terrones 2014
EXCITON	literature	Henriquez-Guerra et al., ACS AMI 15, 57369 (2023)
DOPING_MoS2	literature	Chakraborty et al., Phys. Rev. B 85, 161403 (2012)
RAMAN_CAL	literature	Mignuzzi et al., Phys. Rev. B 91, 195411 (2015)
gamma_E, gamma_A, gamma_LA, C2D, dgap_deps	this work	frozen phonons and biaxial strain sweep, this work
C_A (WS2, MoSe2, WSe2)	this work	predicted in this work by transfer
GAMMA_MEAS	literature	Michail et al., ACS AMI 16, 49602 (2024), MoS2 and WSe2 only
CHL_DOP	literature	Chernikov et al., Phys. Rev. Lett. 115, 126802 (2015)
DEQK_DEPS, DEGK_DEPS	not used	strain-tuned valley shift, zero strain in every reported result
COX, STACK	geometry	nominal gate-stack capacitance and permittivity
VSAT	literature	Smithe et al., Nano Lett. 18, 4516 (2018), MoS2 value used for all four
P_SPEC	assumed	fully diffuse etched edge, $p = 0$
halo_contrast	assumed	20x defect density inside the damage halo
Udef	calibrated once	fixed once on Dossena et al., npj 2D Mater. Appl. 9, 67 (2025)
N_IT	calibrated once	fixed once on measured monolayer field-effect mobility

and the drain current per unit width follows from

$$\frac{I_{\text{on}}}{W} = qn_{\text{s}} \frac{\mu \mathcal{E}}{1 + \mu \mathcal{E}/v_{\text{sat}}}, \quad n_{\text{s}} = \frac{C_{\text{ox}}(V_{\text{ov}} - \Delta V_{\text{T}})}{q}, \quad (\text{S8})$$

with $\mathcal{E} = V_{\text{DS}}/L_{\text{ch}}$. Reported results use $V_{\text{ov}} = 1.5$ V, $V_{\text{DS}} = 1$ V and $L_{\text{ch}} = 300$ nm, matching the measured devices, and the measured monolayer MoS₂ saturation velocity $v_{\text{sat}} = 3.4 \times 10^6$ cm/s.¹⁸ No comparable measurement exists for WS₂, MoSe₂ or WSe₂, so the same value is used for all four rather than a guessed material dependence. The critical width W_{c} is the width at which I_{on}/W falls to half of its wide-channel value at the stated gate overdrive; Fig. S3(c) shows that it grows smoothly, by a factor of 3.8 for a 5 nm halo and by 2.4 for a 150 nm halo, as the criterion is tightened from 0.3 to 0.8, with the ordering of the halo widths unchanged, so the choice of 0.5 fixes the scale of W_{c} but not the comparisons drawn from it. Because the edge-charge term enters through $V_{\text{ov}} - \Delta V_{\text{T}}$, that branch of W_{c} scales inversely with V_{ov} ,

whereas the halo branch does not; the two contributions are therefore reported separately in the main text.

S4. SELF-CONSISTENT DEVICE SOLVE

Every current that the article compares with measurement comes from a self-consistent solve of the channel rather than from the compact expression of Eq. (S8); the design map and growth-technology lines of Fig. 4(b), the four-material screening of Fig. 4(c) and the critical-width scans use the compact expression, which is validated against the self-consistent solve at the end of this section. Two fields are carried at each point along the channel: the surface band bending ψ , that is the local shift of the conduction band edge with respect to the source Fermi level, and the electron quasi-Fermi potential V . They satisfy the vertical electrostatic balance

$$C_{\text{ox}}(V_{\text{ov}} - \Delta V_{\text{T}} - \psi) = qn_{\text{s}}(\psi, V), \quad (\text{S9})$$

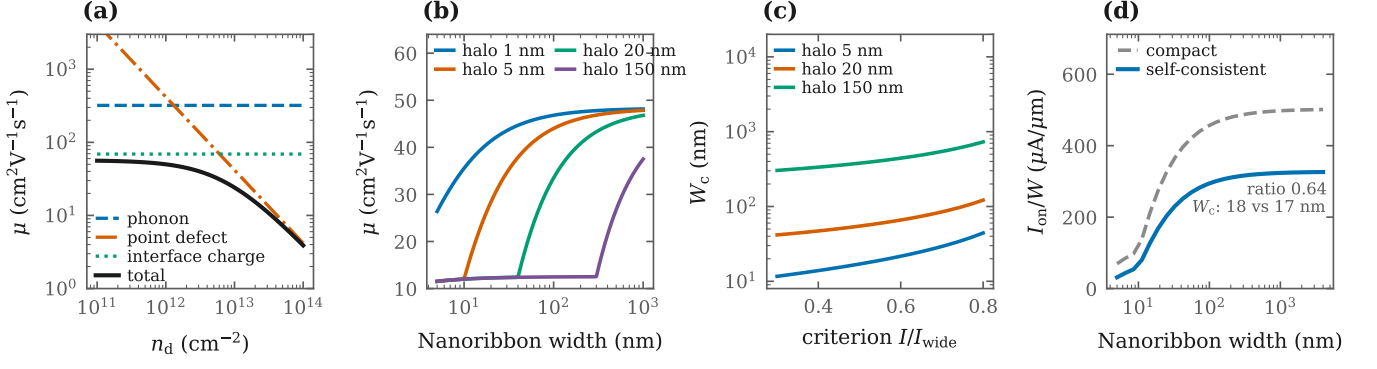


FIG. S3. (a) Decomposition of the MoS₂ sheet mobility into its three scattering channels. (b) Ribbon mobility against width for four damage-halo widths. (c) Sensitivity of the critical width to the choice of criterion. (d) The compact expression of Eq. (S8) against the self-consistent solve of Eqs. (S9) to (S11); the two differ by a nearly constant factor, so the critical width, which is a ratio, is almost unchanged.

with the sheet density of a two-dimensional parabolic band,

$$n_s = g_{2D} k_B T \ln \left[1 + e^{q(\psi-V)/k_B T} \right], \quad g_{2D} = \frac{g_s g_v m^*}{2\pi \hbar^2}, \quad (\text{S10})$$

and current continuity for drift and diffusion,

$$\frac{d}{dx} \left[\frac{qn_s \mu dV/dx}{1 + (\mu/v_{sat})|dV/dx|} \right] = 0. \quad (\text{S11})$$

Writing the charge as Eq. (S10) rather than as $C_{ox}(V_{GS} - V_T)/q$ places the quantum capacitance $C_q = q^2 g_{2D} f(\eta)$ in series with the oxide. The valley degeneracy is $g_v = 2$ for the two K valleys of both bands. The spin degeneracy is $g_s = 2$ for electrons, whose conduction-band spin-orbit splitting at K is a few tens of meV, and $g_s = 1$ for holes, whose valence-band splitting is several hundred meV, so that only the upper spin branch is occupied at the densities of interest.¹⁹ For MoS₂ on the high- κ stack $C_q = 68 \mu\text{F}/\text{cm}^2$, so the series combination removes 3.3% of the charge that the oxide alone would imply.

Because V is the quasi-Fermi potential, drift and diffusion are both carried by its gradient, and the fall of n_s towards the drain that the compact expression omits appears automatically. The two equations are assembled on a one-dimensional finite-volume mesh and solved together by Newton's method in DEVSIM, an open-source device simulator, with the Jacobian formed from the symbolic derivatives of the models.²⁰ The contact resistance is closed by bisection on the internal drain bias: the channel current is monotone in that bias, so $g(v) = v + 2I(v)R_c - V_{DS}$ has a single root, with R_c quoted per contact, whereas a fixed point on the current diverges once the contact carries most of the applied bias, which is exactly the p-type regime of interest.

Four checks run with the code. The band bending returned by the coupled Newton solve matches an independent bisection solution of Eq. (S9) at every node to

1.8×10^{-16} , using only the converged quasi-Fermi potential as input; the residual of Eq. (S9) itself, measured against the gate charge $C_{ox}V_{ov}$, is 1.2×10^{-15} ; the current entering the source equals the current leaving the drain to 2.7×10^{-15} ; and in the long-channel, low-field limit the solver reproduces the textbook square law to 0.2% once C_q is placed in series with the oxide, while the uncorrected square law is high by the 3.3% quoted above.

Figure S3(d) compares the two models over the whole width range. The compact expression is high by a nearly constant factor: the ratio of the self-consistent to the compact current is 0.64 with a spread of 0.05 for $W > 12$ nm, close to the triode factor $(V_{ov} - V_{DS}/2)/V_{ov} = 0.67$ that the compact form leaves out. Because the critical width is defined as a ratio of currents, that common factor cancels almost entirely: $W_c = 18$ nm from the self-consistent solve against 17 nm from the compact expression, a difference of 7%. Every conclusion in the article about how narrow a ribbon can usefully be made therefore rests on a quantity that the two models agree on, while the absolute currents compared with measurement are the self-consistent ones.

S5. PHYSICS-GUIDED ADJOINT INVERSION

The disorder-activated intensity ratio and the A-exciton energy add two further observables, so that a full hyperspectral acquisition constrains the three-component latent state $\theta = (\log_{10} n_d, \varepsilon, n)$ through

$$\omega_i = \omega_i^0 (1 - 2\gamma_i \varepsilon/100) + \lambda_i n, \quad i \in \{E', A'_1\}, \quad (\text{S12})$$

$$R_A = C_A n_d \times 10^{-14}, \quad (\text{S13})$$

$$E_A = E_A^0 + \Xi \varepsilon/1000 + \chi n/1000. \quad (\text{S14})$$

Here χ is the shift of the A-exciton resonance with electron density, for which we take the scale reported for electrostatically gated monolayer WS₂,²¹ where screening of the exciton binding energy outweighs the bandgap

renormalisation. This coefficient enters only the three-field image inversion demonstrated in this section; no result reported in the main article depends on it. For an image we minimise

$$J = \frac{1}{2} \sum_p \|\mathbf{r}_p\|_{C_d^{-1}}^2 + \frac{1}{2} \sum_p \|\Delta\boldsymbol{\theta}_p\|_{C_p^{-1}}^2 + \frac{1}{2} \sum_k \alpha_k \|\nabla\Theta_k\|^2, \quad (\text{S15})$$

where the gradient-squared term encodes the expectation that the three fields are continuous on the scale of the optical resolution and is what couples the pixels. Because the forward operator is an explicit analytic chain, its linearisation G_p is available in closed form and the gradient is assembled in one adjoint sweep,^{22,23}

$$\nabla_p J = G_p^\top C_d^{-1} \mathbf{r}_p + C_p^{-1} \Delta\boldsymbol{\theta}_p - \alpha_k \nabla^2 \Theta_k|_p. \quad (\text{S16})$$

The Gauss-Newton system is solved matrix free by conjugate gradients, so the cost per iteration is one forward and one adjoint application regardless of image size, and the posterior covariance follows from the Laplace approximation $C_{\text{post}} = (G^\top C_d^{-1} G + C_p^{-1})^{-1}$.

Three checks run automatically with the code. The analytic Jacobian matches a central finite difference to a relative error of 1.8×10^{-8} ; the separately implemented forward and adjoint applications satisfy $\langle G\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, G^\top \mathbf{w} \rangle$ to a relative error of 2.1×10^{-16} , that is to machine precision; and the assembled map gradient matches a directional finite difference of the full objective, regulariser included, to 9.3×10^{-10} .

Figure S4 reports the conditioning and a synthetic recovery test on a 48×48 pixel map containing a strained grain boundary, a doped patch and a defective region. To avoid an inverse crime the data are generated with coefficients that differ from those used in the inversion by three percent and with heavy-tailed (Student t) noise, while the inversion assumes Gaussian noise and the unperturbed coefficients. The objective stops decreasing after 4 Gauss-Newton iterations out of the 15 requested, and the solver separates the three fields with root-mean-square errors of 0.095 dex, 0.016% and $0.059 \times 10^{13} \text{ cm}^{-2}$. Removing the intensity-ratio channel leaves the defect density on its prior, which is why an intensity channel is indispensable. Note that the two frequency channels of the image model are the two first-order modes E' and A'_1 : the overtone used in Section S2 is a single-point observable and does not enter here.

S6. INVERSION OF PUBLISHED FILMS

Table S2 lists the inverted state for the four atomic-layer-deposited and sputtered MoS_2 films of Ref. 24, obtained from $I_{\text{LA}}/I_{A'_1} = C_A n_d$ with the MoS_2 calibration constant.⁸ Two disorder-activated ratios were measured independently on the same films; if both are linear in n_d their ratio must be constant, and we find 6.5 ± 1.1 . The residual scatter is systematic rather than random, the

two deposition families clustering at 7.4 and 5.6, which is expected because the shoulder feature and the LA(M) peak have different resonance conditions and the films differ in thickness. The detuning Δ that controls the transfer of C_A across the family is itself observable: biaxial strain shifts the WS_2 B exciton and measurably collapses the double-resonant 2LA(M) intensity, which is direct evidence that the double-resonance picture used here is the right one for this mode.²⁵ An independent calibration on helium-ion irradiated material reports a larger constant,²⁶ reflecting a different defect species; adopting it would rescale all inverted defect densities by a common factor without changing any relative conclusion. As a further check on the forward model, the 50 meV blue shift of the A exciton measured at a WS_2 grain boundary by wide-field hyperspectral reflectance²⁷ corresponds to 0.39% of compression when converted with the measured biaxial gauge factor,²⁸ against the 0.6% quoted in the original work with a bulk-like deformation potential.

The measured mobilities of these films fall between 74 and 479 times below the point-defect ceiling. They are polycrystalline few-layer films whose electron micrographs show vertical domains and grain boundaries,^{24,29} so the deficit is expected and is itself informative: it separates the point-defect contribution, which the optical ratio measures, from the grain-boundary contribution, which it does not.

S7. LIMITATIONS

1. PBE without spin-orbit coupling. Absolute gaps and valley separations carry the known functional spread, so published spin-orbit-resolved effective masses are used^{15,19} and only strain derivatives are taken from our own calculations.
2. The edge charge is inferred from one published data set on one material, and assumes the extracted density is uniform over an edge width of 10 nm.
3. The predicted calibration constants for WS_2 , MoSe_2 and WSe_2 rest on a double-resonance scaling argument and have not been measured.
4. The model treats isolated point defects; grain-boundary networks and dielectric disorder are not resolved, so the mobility ceiling is an upper bound for nanocrystalline films.
5. The transport layer uses a constant relaxation time within each scattering channel, so mobility is evaluated at a single carrier energy rather than from a full energy-resolved integral. The device solve is semiclassical drift and diffusion with a field-dependent velocity, so ballistic and tunnelling transport are outside its scope, and the contact is represented by a lumped series resistance rather than by a resolved Schottky barrier.

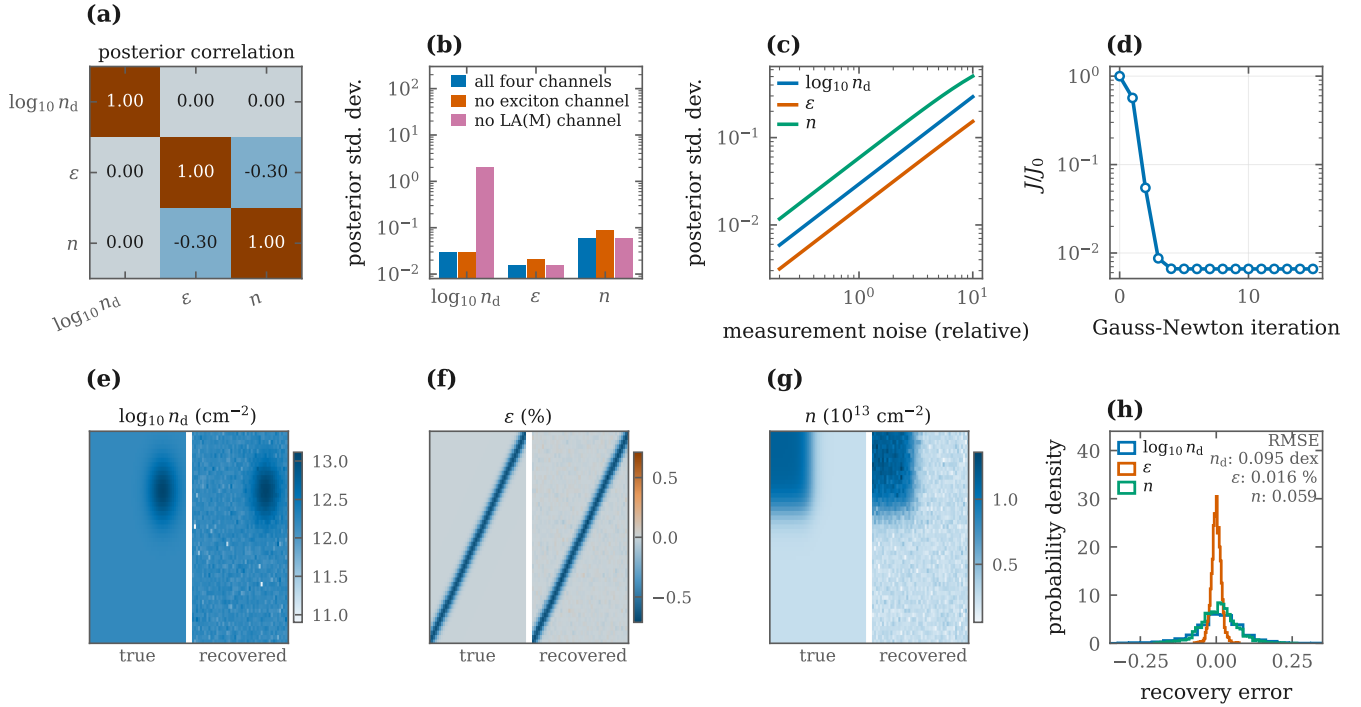


FIG. S4. Adjoint inversion. (a) Posterior correlation matrix. (b) Posterior widths for three observable subsets. (c) Posterior widths against measurement noise. (d) Objective decay of the matrix-free solver. (e)-(g) True and recovered fields. (h) Recovery error distributions.

- The comparison with the first-principles disordered WS $_2$ result¹⁶ is good at 2×10^{13} cm $^{-2}$ (52 against 56.5 cm 2 V $^{-1}$ s $^{-1}$) but the model overestimates at 4×10^{13} cm $^{-2}$ (27 against 8.5), where the mean free path approaches the inter-defect spacing and a semiclassical treatment is no longer adequate.

S8. DATA AND CODE AVAILABILITY

The archive of record is deposited at Zenodo, <https://doi.org/10.5281/zenodo.21670316>.³⁰ It holds the raw PySCF output for every stage described in Section S1, the convergence and k -mesh cross-check runs, the reduced constants, every digitised published measurement used in this work as a flat file with its source stated in the header, the machine-readable results file behind every number quoted here, and the figures as published.

The development repository is <https://github.com/Tanvir-Mahmud-Mahim/Width-scaling-in-monolayer-semiconductor-nanoribbon-transistors>. `dft/` contains the PySCF drivers and the post-processing that reduces raw output to the constants used by the models; `rapid/` contains the parameter table with a provenance tag on every entry, the forward model and its analytic Jacobian, the adjoint solvers and their three verification tests, the transport and nanoribbon layers, the published data sets with their sources, and the scripts that regenerate every number, figure and table. Run-

ning `python -m rapid.analysis`, then `rapid.figures`, `rapid.supplement` and `rapid.numbers` reproduces the article end to end from the stored first-principles output; re-running the density functional theory requires only PySCF and takes a few hours on two cores.

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